

Rathishan Mahendran

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Executive Summary

Computer Engineering undergraduate (2023–2027) at the University of Ruhuna with strong expertise in software engineering, full-stack development, and AI-driven systems. Proficient in Machine Learning, Generative AI, and RESTful API development, with experience building scalable web applications. Strong foundation in data structures and algorithms with a focus on performance, scalability, and user-centric design. Seeking a Software Engineer Internship to develop efficient, scalable, and impactful software solutions.

Technical Skills

- Programming Languages: Python, TypeScript, JavaScript (ES6+), C++
- Backend Development: FastAPI, RESTful API Development, Asynchronous Programming, API Security
- Frontend Development: React, Vue.js, HTML5, CSS3, Tailwind CSS
- Databases: MySQL, PostgreSQL, MongoDB
- AI & Machine Learning: Machine Learning, Generative AI, NLP, Scikit-Learn, LangChain, Prompt Engineering
- DevOps & Tools: Git, GitHub, Docker, CI/CD (GitHub Actions)
- Design & Prototyping: Figma, Framer Motion, VS Code

Soft Skills

- Communication & Teamwork — collaborative in cross-functional academic and project settings
- Problem Solving — strong analytical thinking developed through competitive programming
- Time Management — balancing academic workload, personal projects, and volunteer commitments
- Adaptability — quickly learns and applies new frameworks and tools across diverse projects
- Multilingual Abilities — English (Professional working proficiency), Tamil (Native proficiency)

Professional Experience

Innovation Club, University of Ruhuna

Jan 2026 – Present

Role: Member — AI & Data Science Section

Responsibilities:

- Analyzed AI/ML research papers and created concise technical summaries for the club knowledge base.
- Collaborated in team-based AI projects and participated in cross-disciplinary research discussions.
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Role: ICT Volunteer & Volunteer Teacher

Responsibilities:

- Delivered structured ICT training sessions covering digital literacy and computer fundamentals.
- Prepared lesson materials and exercises tailored to students with varying technical backgrounds.
- Maintained and troubleshooted computer systems, diagnosing hardware and software issues.
- Coordinated session logistics with school administrators and Ministry of Education representatives.
- Provided one-on-one support to students struggling with ICT concepts.

Impact:

- Trained 30+ students across multiple sessions, measurably improving digital literacy outcomes.
- Resolved technical faults across school computer labs, restoring full functionality.

Featured Projects

Personalized Study Buddy — AI Study Assistant

CodeBlast 2025 — Top 10

Stack: Python · FastAPI · MongoDB · Groq API · LangChain · Scikit-Learn

Responsibilities:

- Built scalable FastAPI backend with RESTful APIs supporting personalized learning workflows.
- Designed and optimised MongoDB schema for efficient storage and retrieval of user learning data.
- Implemented ML-based content recommendation system using Scikit-Learn.
- Integrated LLM-powered natural language interaction via Groq API and LangChain.
- Developed asynchronous API endpoints for real-time AI response generation.

Impact:

- Reduced MongoDB query response time by 30% through schema optimisation.
- Achieved Top 10 ranking at CodeBlast 2025 (University of Ruhuna × Mobitel).

ShopNet — Full-Stack E-Commerce Application

Stack: Vue 3 · TypeScript · Pinia · Tailwind CSS v4 · DummyJSON REST API · Netlify

Responsibilities:

- Developed a full-stack SPA with REST API integration and state management via Pinia.
- Implemented cart, user authentication, and checkout systems with TypeScript strict typing.
- Designed a fully responsive UI using Tailwind CSS v4 with mobile-first principles.
- Set up CI/CD pipeline and deployed the application on Netlify.
- Debugged and resolved runtime and TypeScript errors to improve application stability.

Impact:

- Delivered a production-ready e-commerce application with end-to-end feature coverage.
- Maintained zero TypeScript compilation errors through strict typing practices.

Personal Portfolio Website

Stack: React 19 · TypeScript · Tailwind CSS v4 · Framer Motion · Vite

Responsibilities:

- Developed a responsive portfolio site with modern UI/UX and smooth Framer Motion animations.
- Implemented component-based architecture with TypeScript and clean code practices.
- Optimised frontend performance and build speed using Vite tooling.

Impact:

- serves as a professional showcase for recruiters and collaborators.

Education

B.Sc. in Computer Engineering

2023 – 2028 (Expected)

University of Ruhuna — Faculty of Engineering, Hapugala, Galle, Sri Lanka

Specialization: AI Agents, DevOps Engineering, and Software Engineering

Awards & Accomplishments

- Top 10 Finalist — CodeBlast 2025, University of Ruhuna × Mobitel
- Docker Foundations Professional Certificate — Docker, Inc. · 2026
- Generative AI Chatbot Bootcamp — DevTown · 2026
- AI/ML Engineer Stage 1 Certification — SLIIT · 2025
- Generative AI Handbook — Educative.io · 2025
- Become an AI Engineer Learning Path — LinkedIn Learning · Ongoing
- IEEE Student Branch Member — University of Ruhuna